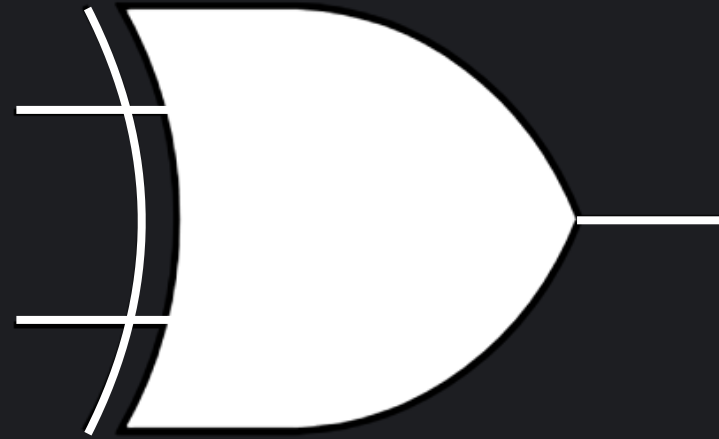


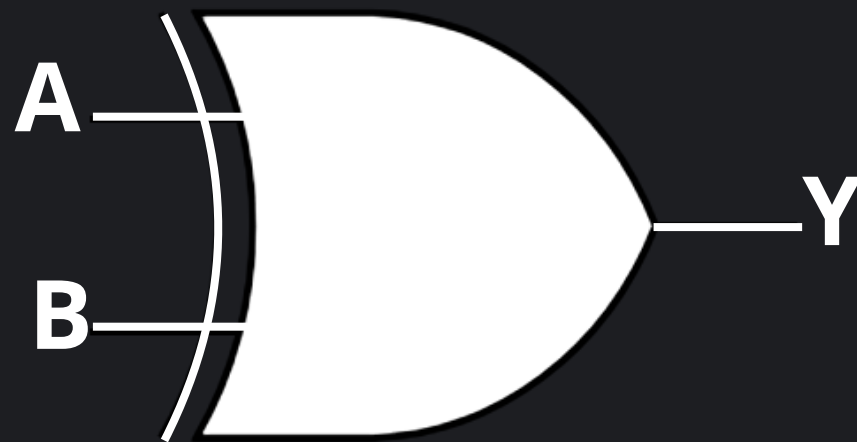
XOR GATE



What is a XOR Gate?

An XOR gate is a basic logic gate that produces a HIGH (1) output when its inputs are different. The output is LOW (0) when all inputs are the same (either all LOW or all HIGH).

Truth Table



A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Function of the XOR Gate

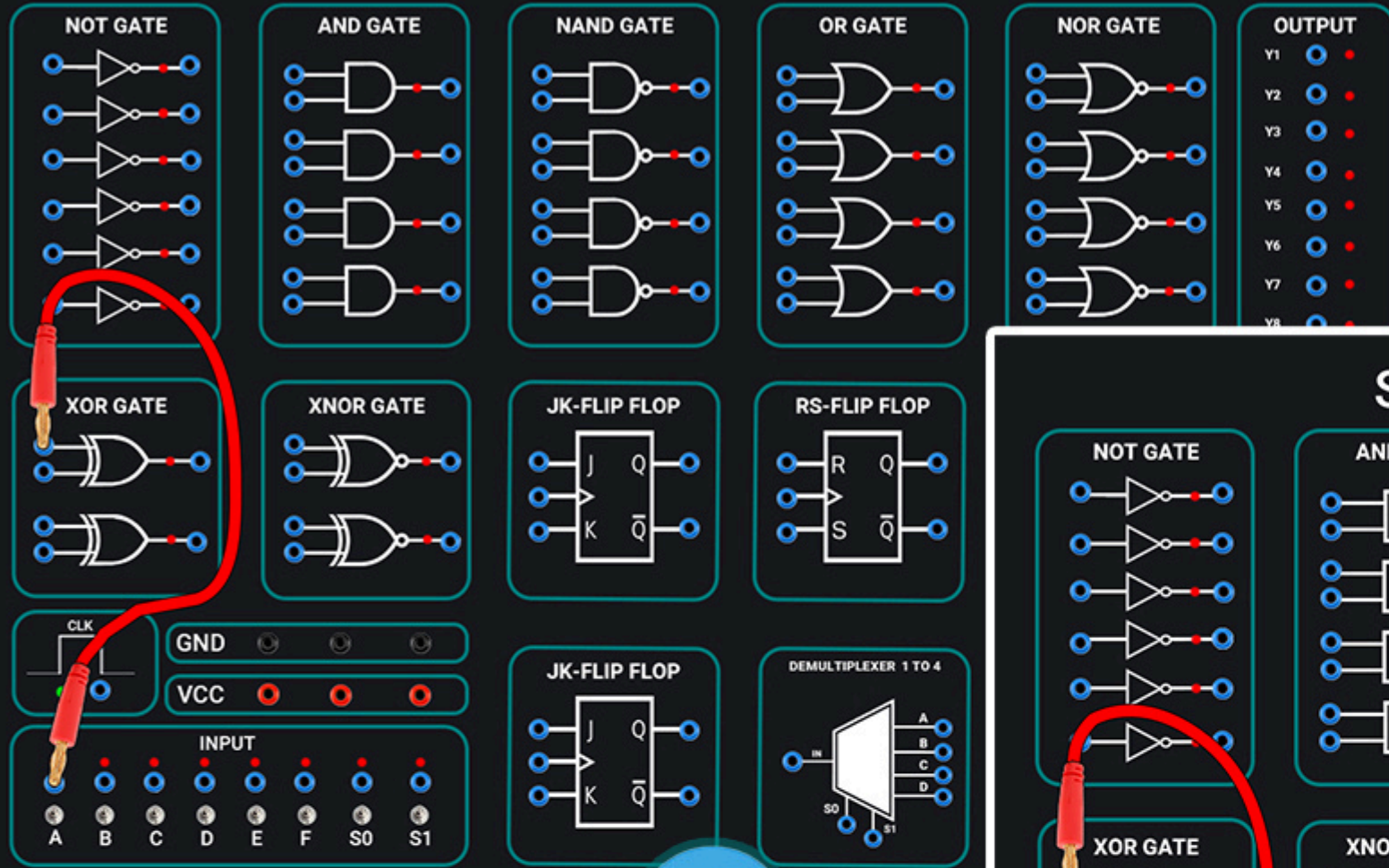
An XOR gate performs a logical Exclusive OR (XOR) operation on its inputs. The main function of an XOR gate is to combine two or more input signals and produce an output signal according to this rule:

Output = 1 when the inputs are different

Output = 0 when the inputs are the same (either all 0 or all 1)

-

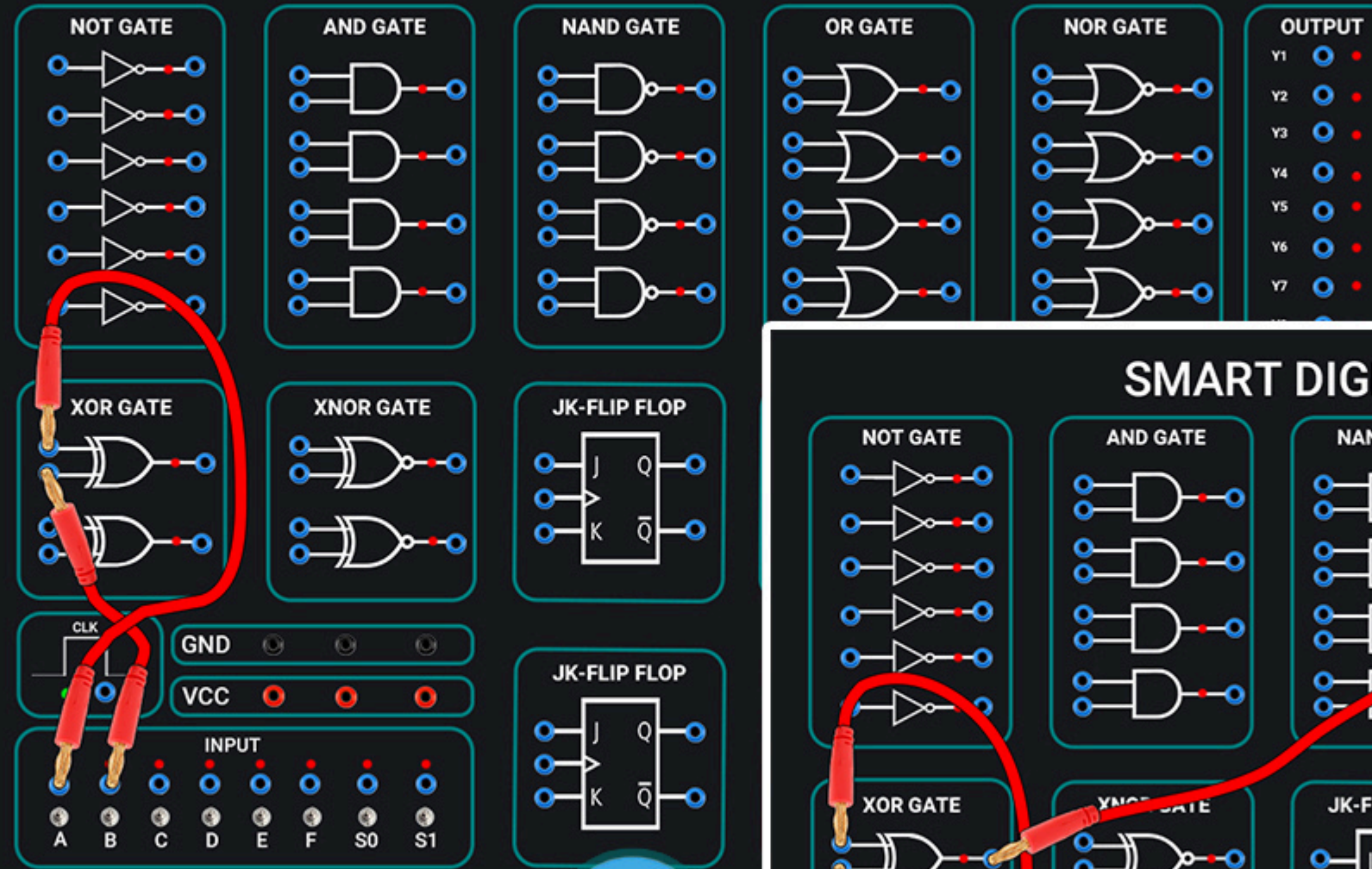
SMART DIGITAL CIRCUITS BOARD



Connect a wire from the first input (top blue terminal) to switch A in the INPUT section.

1

SMART DIGITAL CIRCUITS BOARD



Connect a wire from the second input (bottom blue terminal) to switch B in the INPUT section

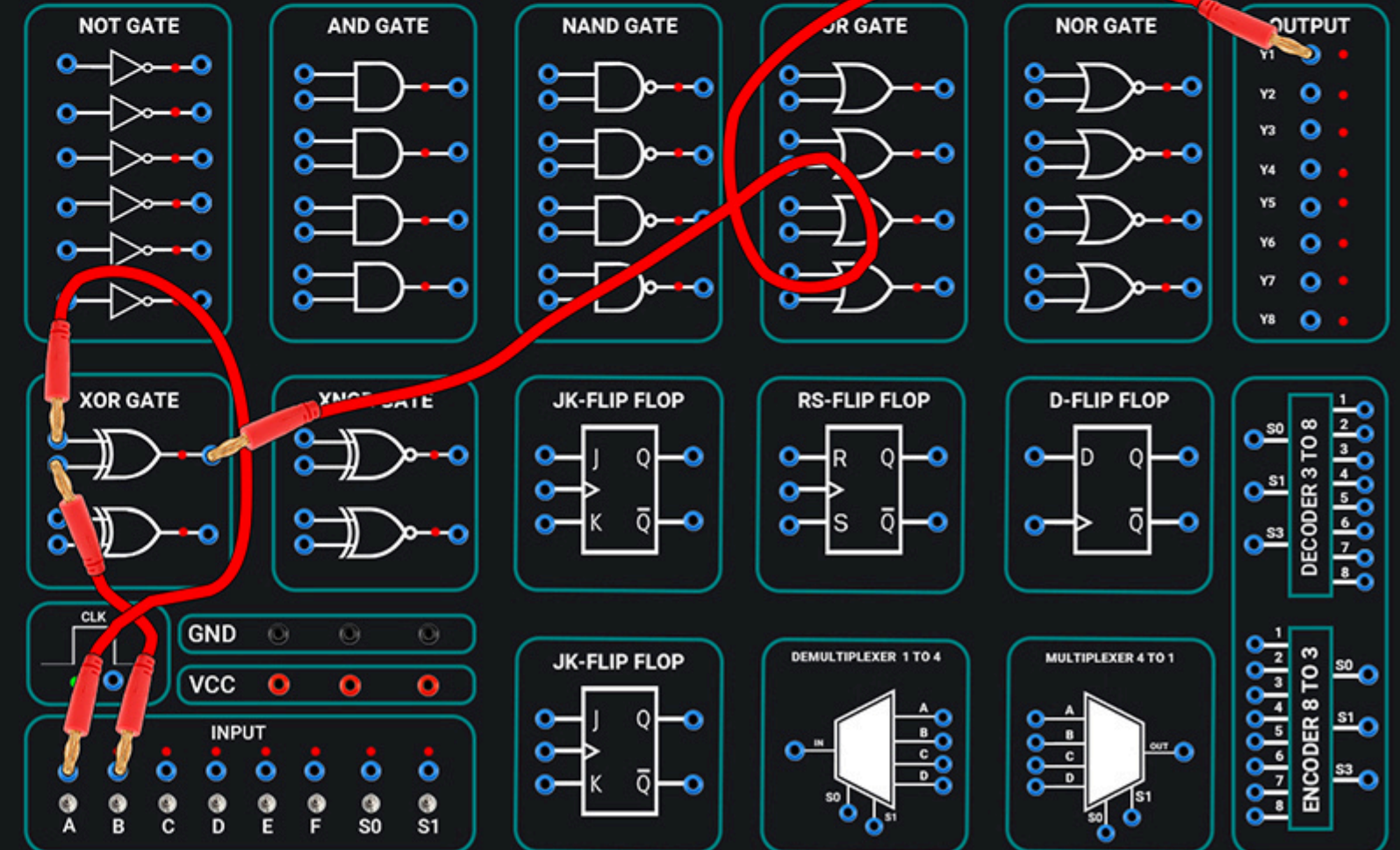
2

Connect the XOR Gate

Connect a wire from the gate output to terminal Y1 (or any point from Y1 to Y8) in the OUTPUT section at the top right.

3

SMART DIGITAL CIRCUITS BOARD



SMART DIGITAL CIRCUITS BOARD

NOT GATE

AND GATE

NAND GATE

OR GATE

NOR GATE

OUTPUT

Y1

Y2

Y3

Y4

Y5

Y6

Y7

Y8

XOR GATE

XNOR GATE

JK-FLIP FLOP

RS-FLIP FLOP

D-FLIP FLOP

DECODE 3 TO 8

1

2

3

4

5

6

7

8

CLK

GND

VCC

INPUT

A

B

C

D

E

F

S0

S1

JK-FLIP FLOP

DEMUX 1 TO 4

MUX 4 TO 1

DECODE 8 TO 3

1

2

3

4

5

6

7

8

S0

S1

S3

